

4.  ~~$E(X_i - \mu)^2$~~  i.i.d,  $E(X_i - \mu)^2 = \sigma^2$

$$\therefore \frac{\sum_{i=1}^n (X_i - \mu)^2}{n} \xrightarrow{P} \sigma^2$$

5. 不能假设  $E X_i$  都为 0, 否则用  $X_i - E X_i$  代替  $X_i$  不是原题了.

11  ~~$r(X_i, Y_j) = \frac{E X_i Y_j - E X_i E Y_j}{\sqrt{E X_i^2} \sqrt{E Y_j^2}}$~~

12 由 Chebyshev 不等式,  $P\left(\left|\frac{1}{n} \sum_{k=1}^n X_k - \frac{1}{n} \sum_{k=1}^n E X_k\right| \geq \varepsilon\right) \leq \frac{D \frac{\sum_{k=1}^n X_k}{n}}{\varepsilon^2} = \frac{D \sum_{k=1}^n X_k}{n^2 \varepsilon^2}$

只需证  $\lim_{n \rightarrow \infty} \frac{D \sum_{k=1}^n X_k}{n^2} = 0$

$$\text{Cov}(X_i, X_j) = \sqrt{D X_i} \sqrt{D Y_j} r(X_i, X_j) \leq C \lim_{|i-j| \rightarrow \infty} \text{Cov}(X_i, X_j) = 0$$

$$\frac{1}{n^2} D \sum_{k=1}^n X_k = \frac{1}{n^2} \left( \sum_{k=1}^n D X_k + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right) \quad \frac{1}{n^2} \sum_{k=1}^n D X_k \leq \frac{C n}{n^2} \rightarrow 0$$

又  $\forall \delta > 0$ ,  $\exists N$ , 当  $|i-j| \geq N$  时,  $|\text{Cov}(X_i, X_j)| < \delta$

$$\therefore \text{又对 } n > \frac{2}{\delta}, \quad \frac{1}{n^2} \sum_{i \neq j} \text{Cov}(X_i, X_j) = \frac{1}{n^2} \left( \sum_{|i-j| \leq N} \text{Cov}(X_i, X_j) + \sum_{|i-j| > N} \text{Cov}(X_i, X_j) \right)$$

$$\leq \frac{1}{n^2} (2nN - N^2) C + (n(n-1) - (2nN - N^2)) \delta$$

令  $n \rightarrow +\infty$ , 由  $\delta$  的任意性, 得该式趋近于 0

原式得证

9. 设  $X_i = \cos(iX)$

则  $EX_i = 0, DX_i = \frac{1}{n} \quad \text{且} \quad \text{Cov}(X_i, X_j) = 0$

由第5题结论可知  $\frac{1}{n} \sum_{k=1}^n X_k - \frac{1}{n} \sum_{k=1}^n EX_k \xrightarrow{P} 0$

$\therefore \frac{S_n}{n} \xrightarrow{P} 0$

11. 设  $X_i = \begin{cases} 1 & \text{若第 } i \text{ 个球在 } A \text{ 袋中} \\ 0 & \text{其他} \end{cases}$

则  $P_{X_i} \sim \begin{pmatrix} \frac{1}{n} & 0 \\ \frac{1}{n} & 1-\frac{1}{n} \end{pmatrix} \quad S_n = \sum_{i=1}^n X_i$

$EX_i = \frac{1}{n} \quad DX_i = \frac{n-1}{n^2}$

$\text{Cov}(X_i, X_j) = \frac{1}{n^2(n-1)}$

$\therefore DS_n = D \sum_{i=1}^n X_i = \sum_{i=1}^n DX_i + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j) = 1$

$\therefore P\left(\left|\frac{S_n - ES_n}{n}\right| \geq \epsilon\right) \leq \frac{P\left(\frac{S_n}{n}\right)}{\epsilon^2} = \frac{1}{n^2 \epsilon^2} \rightarrow 0$

$\therefore \frac{S_n - ES_n}{n} \xrightarrow{P} 0$

$$4. E\bar{X} = E \frac{X_1 + \dots + X_n}{n} = E X_1 = 1$$

$$\text{Var}(\bar{X}) = \text{Var} \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n^2} \cdot n \text{Var} X_1 = \frac{1}{3n}$$

10.

$$\sum_{i,j} (X_i - X_j)^2 = (n-1) \sum_{i=1}^n X_i^2 - 2 \sum_{i < j} X_i X_j$$

$$= n \sum_{i=1}^n X_i^2 - \left( \sum_{i=1}^n X_i \right)^2$$

$$= n \sum_{i=1}^n (X_i - \bar{X})^2$$

$$\therefore \frac{1}{n(n-1)} \sum_{i,j} (X_i - X_j)^2 = S^2$$

24. ~~24.1~~

$$(1) P(X_{(6)} > 10) = 1 - P(X_{(6)} \leq 10) = 1 - (P(X_1 \leq 10))^{16} = 1 - (\Phi(1))^{16}$$

$$(2) P(X_{(6)} > 5) = (P(X_1 > 5))^{16} = (1 - \Phi(-1.5))^{16} = (\Phi(1.5))^{16}$$

$$29. (1) f_{F(X)}(\eta) = \begin{cases} 1 & 0 \leq \eta \leq 1 \\ 0 & \text{其他} \end{cases} \quad \therefore F(X) \sim U(0, 1)$$

$\therefore F(X_{(6)})$  为取自  $U(0, 1)$  的次序统计量, 有  $F(X_{(6)}) \sim \text{Be}(6, 5)$

$$\therefore E(F(X_{(6)})) = \frac{6}{11} \quad \text{Var}(F(X_{(6)})) = \frac{5}{242}$$

(2) 在  $x=0.15$  函数值为  $\int_0^{0.15} \frac{1}{\beta(6,5)} x^5 (1-x)^4 = 0.0014$

32.  $f(x, y) = \frac{5!}{1!1!1!} \cdot x^2 (y^3 - x^3) (1-y^3) \cdot 3x^2 \cdot 3y^2$   
 $= 1080 x^5 y^2 (y^3 - x^3) (1-y^3) \quad 0 < x \leq y < 1$   
 $\chi_{(2)}, \chi_{(4)}$

$f(u, v) = f(uv, v) v = 1080 u^5 (1-u^3) v^2 (1-v^3)$   
 $\frac{\chi_{(2)}}{\chi_{(4)}}, \chi_{(4)}$   
 可分离成  $u$  的系数与  $v$  的系数

$\therefore \frac{\chi_{(2)}}{\chi_{(4)}} \text{ 与 } \chi_{(4)} \text{ 相互独立}$